Extension of specmine and WebSpecmine for Unsupervised Analysis

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Abstract. Metabolomics is a systems biology field focused on the comprehensive identification and quantification of metabolites in biological samples. specmine and WebSpecmine are bioinformatics tools developed by the BiSBi group at CEB, in collaboration with UFSC, for the integrated analysis of metabolomics and spectroscopy data. Despite their scientific impact, the current versions do not incorporate recent advances in unsupervised learning, particularly non-linear dimensionality reduction techniques such as UMAP and t-SNE. This work extends both tools with new dimensionality reduction methods (UMAP, t-SNE, ICA) and clustering algorithms (DBSCAN, HDBSCAN, Gaussian Mixture Models, Hierarchical Clustering), supporting 2D and 3D interactive visualisations, and adds a preprocessing module and an embedding comparison dashboard to WebSpecmine. Validation on five reference datasets demonstrates that the new implementations produce biologically consistent results, with UMAP and t-SNE achieving trustworthiness scores above 0.93 compared to 0.82 for PCA on a Raman spectroscopy dataset.

Keywords: Metabolomics · Unsupervised Learning · Clustering · UMAP · t-SNE · specmine · WebSpecmine

1 Introduction

Metabolomics is a systems biology field that aims at the comprehensive identification and quantification of the metabolites present in a biological sample. To this end, it relies on high-resolution analytical techniques such as Nuclear Magnetic Resonance (NMR), Gas Chromatography coupled to Mass Spectrometry (GC-MS), and Liquid Chromatography coupled to Mass Spectrometry (LC-MS), which enable the simultaneous measurement of hundreds or even thousands of metabolites. The result is high-dimensionality datasets whose manual analysis is unfeasible, making the use of specialised computational tools for their processing and interpretation indispensable.

Metabolomics has emerged as a key driver in precision medicine and systems biology, enabling the identification of disease-associated biomarkers across a wide range of conditions, including cancer, diabetes, cardiovascular disorders, and neurodegenerative diseases [7]. Unlike genomics or transcriptomics, the metabolome is highly sensitive to both genetic and environmental factors, making it a direct readout of an organism's physiological state. The growing integration of metabolomics with other omics layers, namely genomics, transcriptomics, and proteomics, further amplifies its potential for understanding complex biological systems and supporting personalised therapeutic strategies [7].

The processing of these data involves multiple steps, from preprocessing (normalisation, alignment, noise removal) to the application of statistical and machine learning methods, with the goal of identifying relevant patterns, discriminating sample groups, and discovering potential biomarkers. In this context, the BiSBi group at CEB (Centre of Biological Engineering), in collaboration with the Federal University of Santa Catarina (UFSC, Brazil), developed bioinformatics tools dedicated to this analysis. Among these, the R package specmine stands out for the integrated analysis of metabolomics and spectroscopy data [1], alongside WebSpecmine, a web application built on specmine that allows users without programming experience to perform sophisticated analyses through an accessible graphical interface [2].

Despite the maturity of specmine and WebSpecmine, their current implementations do not fully cover recent advances in unsupervised methodologies. This paper describes the extensions developed in this project, contributing: (i) integration of UMAP, t-SNE, and ICA as new dimensionality reduction methods, with support for interactive 2D and 3D visualisations; (ii) integration of DBSCAN, HDBSCAN, Gaussian Mixture Models (GMM), and Hierarchical Clustering (HC) as new clustering algorithms; (iii) a new preprocessing module in WebSpecmine covering data cleaning, imputation, and transformation; (iv) a new embedding comparison dashboard with trustworthiness and continuity metrics; and (v) a PCA/HCA panel with loading plots, heatmaps, dendrograms, and automatic loading interpretation.

2 State of the Art

2.1 Existing Tools for Metabolomics Analysis

Several software platforms are currently available for metabolomics data analysis, each targeting different aspects of the analytical pipeline. Table 1 provides a comparative overview of the most widely used tools, highlighting key differences in scope, interface, and support for modern unsupervised learning methods.

XCMS Online is widely used for preprocessing of LC-MS and GC-MS data, offering feature detection, alignment, and statistical comparison, but focuses almost exclusively on preprocessing and does not include downstream machine learning methods [8]. MZmine provides flexible, modular workflows for mass spectrometry data processing but similarly lacks support for modern dimensionality reduction techniques [8]. MetaboAnalyst is arguably the most feature-complete web-based platform, offering preprocessing, statistical analysis, pathway analysis, and visualisation; however, it does not natively support UMAP or t-SNE [9]. specmine and WebSpecmine distinguish themselves by supporting

Table 1: Comparison of metabolomics analysis platforms.

Tool	**Interface**	**Data Types**	**Clustering**	**PCA**	**UMAP/t-SNE**
specmine	R package	NMR, MS, UV, IR, Raman	Yes	Yes	No
WebSpecmine	Web (GUI)	NMR, MS, UV, IR, Raman	Yes	Yes	No
MetaboAnalyst	Web (GUI)	NMR, MS	Yes	Yes	No
XCMS Online	Web (GUI)	LC/GC-MS	No	Yes	No
MZmine	Desktop	LC/GC-MS	No	Yes	No

multiple spectroscopic data types within a unified environment and by being fully open-source and extensible. As shown in Table 1, no existing platform currently integrates UMAP or t-SNE natively, representing a clear gap that this project addresses.

2.2 Unsupervised Learning in Metabolomics

Unsupervised learning encompasses a set of methods that identify patterns and structure in data without relying on predefined class labels. In metabolomics, these methods are applied in two main categories: **clustering** and **dimensionality reduction**.

Clustering. Clustering methods group samples according to their similarity without prior knowledge of group membership. The main families of algorithms relevant to metabolomics include partitioning methods (e.g., k-means, k-medoids), hierarchical methods (e.g., agglomerative clustering), density-based methods (e.g., DBSCAN, HDBSCAN), and model-based methods (e.g., Gaussian Mixture Models). Density-based approaches are particularly useful in metabolomics as they can detect arbitrarily shaped clusters and identify outliers, which may reflect sample degradation or analytical artefacts.

Dimensionality Reduction. Principal Component Analysis (PCA) remains the most widely used linear method, already implemented in specmine [1]. Independent Component Analysis (ICA) separates multivariate signals into statistically independent components, capturing variation sources beyond the reach of PCA. Non-linear methods such as UMAP and t-SNE have emerged as powerful alternatives, demonstrated to outperform PCA in visualising complex omics data [5, 6]. Table 2 summarises the main methods considered.

2.3 UMAP and t-SNE

UMAP and t-SNE have demonstrated promising results in high-dimensional omics analysis [3, 4]. Both model proximity relationships between points and

Table 2: Comparison of dimensionality reduction methods.

Method	Type	Global Structure	Scalability	In specmine
PCA	Linear	Yes	High	Yes
ICA	Linear	Partial	High	No
t-SNE	Non-linear	No	Low ($O(n^2)$)	No
UMAP	Non-linear	Yes	High	No

project them onto a lower-dimensional representation that preserves local structure. t-SNE is based on conditional probability distributions, while UMAP relies on fuzzy simplicial sets grounded in Riemannian geometry and algebraic topology [3]. Becht et al. [5] showed that UMAP outperforms t-SNE in cluster compactness and global structure preservation, while scaling approximately linearly with dataset size. Kobak and Berens [6] demonstrated that properly configured t-SNE, with PCA initialisation and adaptive learning rates, remains highly competitive with UMAP for single-cell and omics data exploration.

3 Methods

3.1 Extensions to specmine

All new methods were implemented as functions within the specmine R package, following its existing conventions for data structures, parameter handling, and documentation. New functions accept specmine dataset objects as input, ensuring seamless integration with existing preprocessing and analysis pipelines.

Dimensionality Reduction. Three new dimensionality reduction methods were integrated into specmine:

- **UMAP**: implemented via the `uwot` R package, exposing key parameters including the number of neighbours (`n_neighbors`), minimum distance (`min_dist`), number of output dimensions (`n_components`), and distance metric. Both 2D and 3D projections are supported.
- **t-SNE**: implemented via the `Rtsne` package, with PCA initialisation enabled by default following the protocol of Kobak and Berens [6]. Exposed parameters include perplexity, number of iterations, and learning rate. Both 2D and 3D projections are supported.
- **ICA**: implemented via the `fastICA` package, separating spectral data into statistically independent components as a complement to PCA. Both 2D and 3D projections are supported.

Clustering. Four new clustering algorithms were integrated:

- **DBSCAN** and **HDBSCAN**: density-based methods implemented via the `dbscan` R package, capable of detecting non-convex clusters and automatically identifying outlier samples.
- **Gaussian Mixture Models (GMM)**: implemented via the `mclust` package, providing probabilistic cluster assignments with automatic model selection via the Bayesian Information Criterion (BIC).
- **Hierarchical Clustering (HC)**: agglomerative clustering with configurable linkage methods (euclidean, ward, complete, average), producing dendrograms for visual inspection of cluster structure.

All clustering methods operate on the 2D coordinates produced by any of the implemented dimensionality reduction methods (PCA, UMAP, t-SNE, ICA), as well as directly on the original data matrix.

Embedding Quality Metrics. To enable principled comparison of dimensionality reduction methods, trustworthiness [10] and continuity scores were integrated. Trustworthiness measures the degree to which the local neighbourhood structure of each sample in the original high-dimensional space is preserved in the low-dimensional embedding, with values ranging from 0 (poor) to 1 (perfect).

3.2 Extensions to WebSpecmine

The WebSpecmine web application was extended with five new functional modules, accessible through a redesigned navigation interface:

- **Preprocessing tab**: a self-contained data preparation module offering automatic cleaning (*Auto clean*), manual operations (conversion of infinite values, removal of variables or samples with missing values), and a full imputation and transformation panel supporting mean/median/kNN imputation, log transformation, mean centering, scaling, and sum normalisation.
- **Embeddings tab**: an interactive panel exposing all dimensionality reduction methods (PCA, UMAP, t-SNE, ICA) with configurable parameters and toggle between 2D and 3D visualisation, rendered via `plotly` for interactive exploration.
- **Clustering tab**: an interactive panel exposing all clustering algorithms with configurable parameters, applied to embedding coordinates selected by the user.
- **Comparison tab**: a dashboard that computes and displays trustworthiness and continuity scores for all selected embedding methods side by side, allowing quantitative comparison without requiring programming knowledge.
- **PCA/HCA tab**: a dedicated panel for loading analysis, including PCA loading bar charts, a loading heatmap, an HCA dendrogram, an HCA heatmap of the original data, automatic textual interpretation of the top contributing features per principal component, and a searchable table of top bands per PC.

3.3 Datasets

Five datasets were used to validate the implemented methods:

- **Iris**: the classic Fisher iris dataset (150 samples, 4 features, 3 classes), used as a controlled benchmark for embedding and clustering validation.
- **Wine**: a chemical analysis dataset of wines from three Italian cultivars (178 samples, 13 features), used to validate multi-class separation in a metabolomics-like context.
- **Cachexia**: a metabolomics dataset of urine NMR profiles from cancer cachexia patients and controls, provided by the specmine package [1].
- **Propolis**: a spectroscopic dataset of propolis samples of different botanical origins, used to validate the PCA/HCA loading analysis features.
- **Raman**: a Raman spectroscopy dataset with 77 samples and over 1500 spectral features, used as the primary validation dataset for embedding quality comparison.

4 Results

4.1 WebSpecmine Interface

Figure 1 shows the redesigned Load Workspace interface, which supports loading built-in test datasets (iris, wine, cachexia, propolis) for immediate testing, as well as user-supplied datasets in metabolomics or spectroscopy format with optional metadata. The Preprocessing tab (Figure 2) provides a three-panel workflow for data quality control: the Auto clean panel applies a recommended preprocessing sequence in one click, the Manual cleaning panel allows targeted removal of problematic variables or samples, and the Imputation and transforms panel exposes the full set of imputation methods and data transformations.

4.2 Dimensionality Reduction

Figure 3 shows the embedding plots produced for the Raman spectroscopy dataset using PCA, UMAP (2D and 3D), ICA (2D and 3D), and t-SNE (2D and 3D). PCA reveals a broad distribution of samples with some visible grouping structure along the first two principal components. UMAP produces more compact and clearly separated cluster structures, with both the 2D and 3D projections revealing distinct groups of samples that are less apparent in the PCA projection. t-SNE similarly produces well-separated local clusters, particularly visible in 3D. ICA components capture independent sources of spectral variation, providing a complementary view to PCA by revealing non-Gaussian structure in the data.

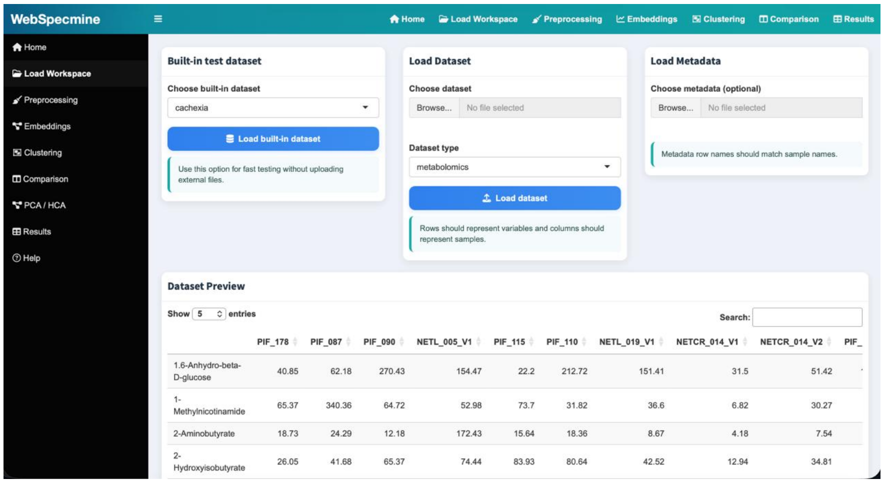


Fig. 1: Load Workspace tab in WebSpecmine, supporting built-in datasets and user-uploaded files.

4.3 Embedding Comparison

Table 3 presents the trustworthiness and continuity scores for PCA, UMAP, and t-SNE computed on the Raman dataset (77 samples, 2 components, $k = 5$ neighbours). Both non-linear methods substantially outperform PCA, confirming that the local neighbourhood structure of the high-dimensional Raman spectra is better preserved by UMAP and t-SNE embeddings. Figure 4 shows the Comparison tab interface.

Table 3: Embedding quality metrics on the Raman spectroscopy dataset ($n = 77$, 2 components, $k = 5$ neighbours).

Method	**Trustworthiness**	**Continuity**
PCA	0.8245	0.8581
UMAP	0.9376	0.9661
t-SNE	0.9638	0.9741

4.4 Clustering

Figure 5 shows clustering results obtained on the Raman dataset using Hierarchical Clustering and HDBSCAN applied to the PCA 2D coordinates. HC identifies a set of well-defined groups of samples, with the cluster boundaries visually consistent with the grouping structure already apparent in the PCA

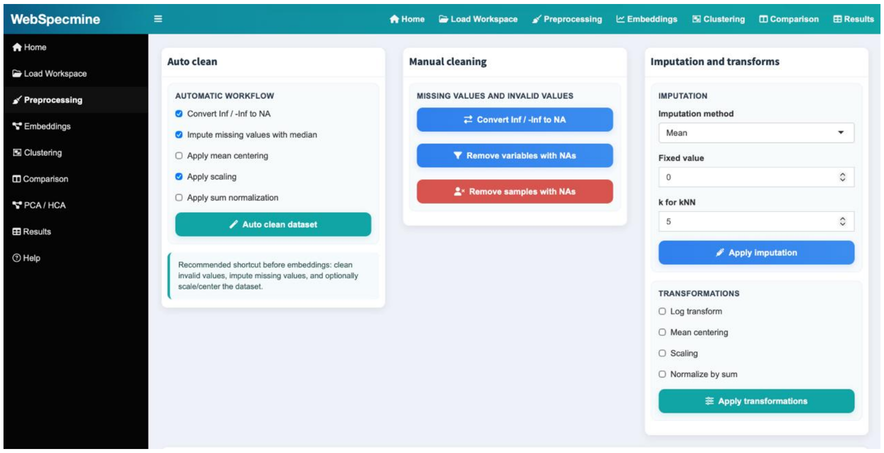


Fig. 2: Preprocessing tab in WebSpecmine, with Auto clean, Manual cleaning, and Imputation and transforms panels.

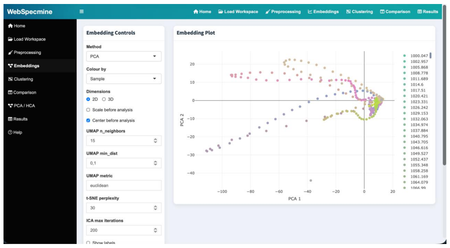
projection. HDBSCAN, operating as a density-based method, identifies a larger number of fine-grained clusters and marks low-density samples as noise (cluster -1), demonstrating its ability to detect outlier samples in spectroscopic data without requiring a predefined number of clusters.

4.5 PCA Loading Analysis and HCA

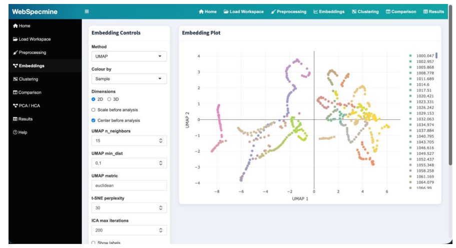
Figure 6 shows the PCA/HCA panel applied to the Raman spectroscopy dataset. The loading bar chart identifies the top contributing spectral bands per principal component, and the automatic interpretation module generates a textual summary of the most important features (e.g., PC1 is mainly driven by bands at 384, 365, and 43 cm^{-1}). The loading heatmap and HCA dendrogram provide a complementary view of the variable structure, revealing groups of co-varying spectral bands. The HCA heatmap of the original data and the Top Bands per PC table with absolute loading values complete the panel, enabling full exploratory analysis of spectral contributions directly in the web interface.

5 Discussion

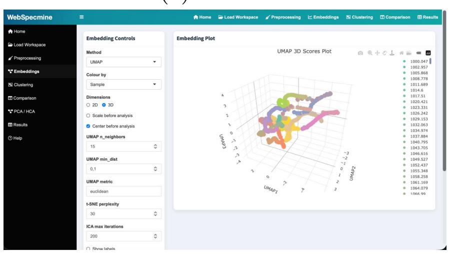
The results demonstrate that the new implementations are functionally correct and produce outputs consistent with the expected behaviour described in the literature. The non-linear dimensionality reduction methods UMAP and t-SNE clearly outperform PCA in terms of trustworthiness and continuity on the Raman spectroscopy dataset, with improvements of approximately 11–14 percentage points in trustworthiness (Table 3). This is consistent with findings in transcriptomics and single-cell biology [5], and confirms that the same benefits



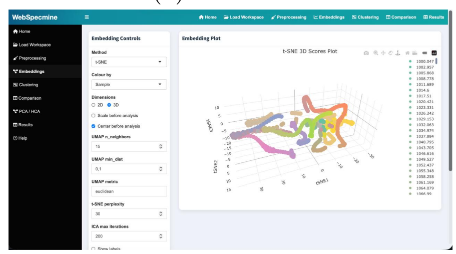
(a) PCA 2D



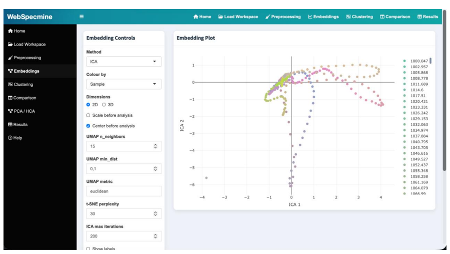
(b) UMAP 2D



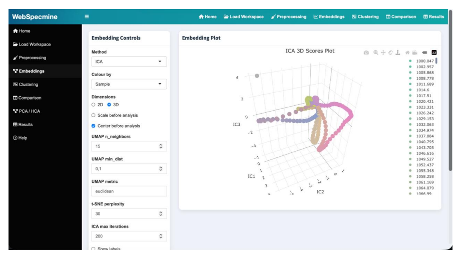
(c) UMAP 3D



(d) t-SNE 3D



(e) ICA 2D



(f) ICA 3D

Fig. 3: Embedding plots for the Raman spectroscopy dataset. All embeddings were computed with centering applied.

of non-linear projection apply to high-dimensional spectroscopic metabolomics data. t-SNE achieves marginally higher scores than UMAP on this dataset, in agreement with Kobak and Berens [6], who showed that properly initialised t-SNE can match or exceed UMAP performance for local structure preservation. UMAP, however, offers faster computation and better global structure preservation, making it more suitable for large-scale datasets.

The clustering results illustrate the complementary nature of the integrated algorithms. HC provides an interpretable, hierarchical view of sample relationships that is particularly useful for exploratory analysis where the number of groups is not known in advance. HDBSCAN offers automatic cluster discovery and outlier identification without requiring a fixed number of clusters, which is a relevant practical advantage for metabolomics datasets where analytical arte-

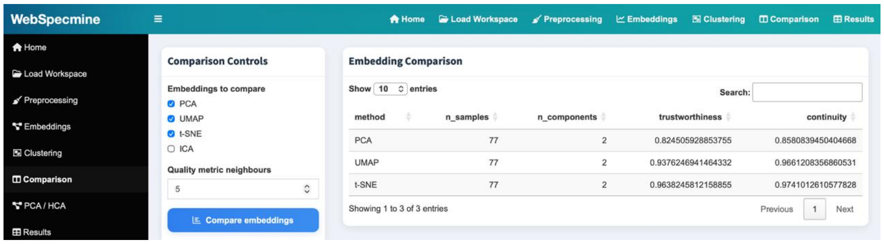
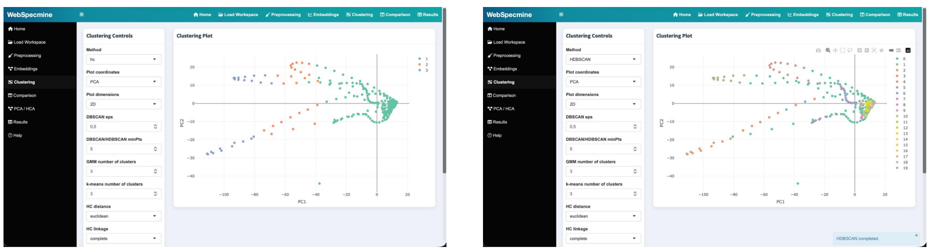


Fig. 4: Comparison tab in WebSpecmine, showing trustworthiness and continuity scores for PCA, UMAP, and t-SNE on the Raman dataset.



(a) Hierarchical Clustering

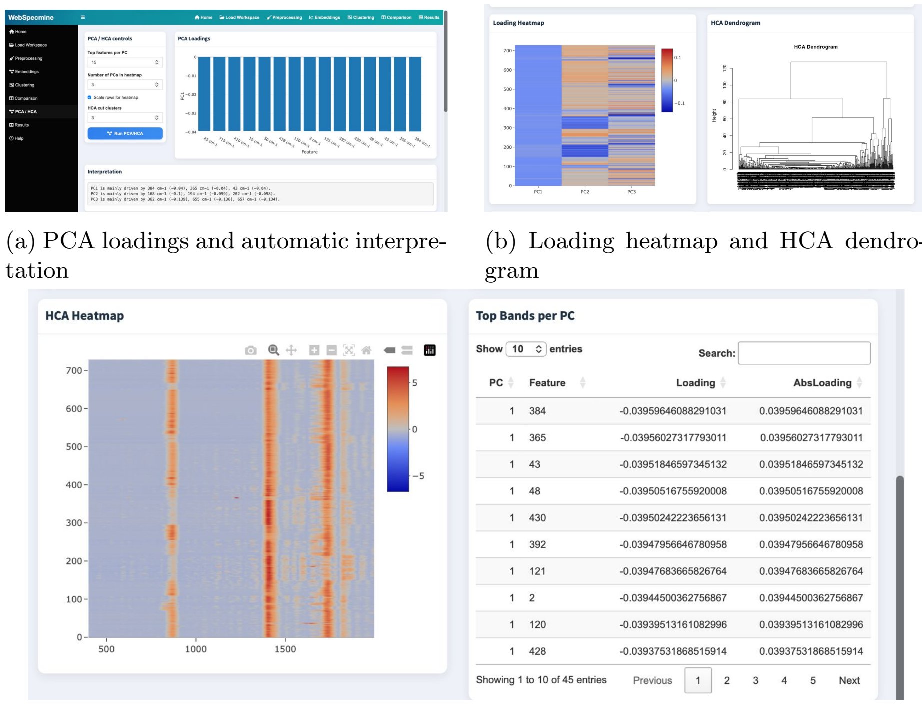
(b) HDBSCAN

Fig. 5: Clustering results on the Raman dataset using PCA 2D coordinates. Colours indicate cluster assignments.

facts may produce isolated samples. The availability of both methods within specmine and WebSpecmine allows users to select the approach best suited to their data and research question.

The PCA/HCA panel, with its automatic loading interpretation, addresses a common bottleneck in spectroscopic analysis: identifying which spectral bands drive the principal components. By surfacing the top features per PC alongside a loading heatmap and dendrogram, WebSpecmine now allows users without R programming experience to perform complete feature-level interpretation of their data. The Comparison tab similarly democratises the evaluation of embedding quality: rather than requiring users to manually compute and compare trustworthiness scores, the interface computes these metrics on demand and presents them in a directly comparable format.

A current limitation is that the 3D visualisations (UMAP 3D, t-SNE 3D, ICA 3D) are rendered interactively in the browser via `plotly` but cannot be exported as static figures directly from the interface. Furthermore, the new WebSpecmine modules currently operate as a standalone extension and are not yet fully merged into the upstream specmine R package; this integration is planned as part of future development. Future work should also address export functionality and extend the Comparison tab to include clustering quality metrics such as the Silhouette coefficient, allowing users to jointly evaluate embedding and clustering quality within a single view.



(a) PCA loadings and automatic interpretation

(b) Loading heatmap and HCA dendrogram

(c) HCA heatmap and Top Bands per PC table

Fig. 6: PCA/HCA panel in WebSpecmine applied to the Raman dataset.

6 Conclusions

This paper described the extension of specmine and WebSpecmine with a comprehensive set of new unsupervised learning capabilities. The implemented contributions include three new dimensionality reduction methods (UMAP, t-SNE, ICA) with 2D and 3D visualisation support, four new clustering algorithms (DBSCAN, HDBSCAN, GMM, HC), a preprocessing module, an embedding comparison dashboard, and a PCA/HCA loading analysis panel. Validation on five reference datasets, including a Raman spectroscopy dataset, confirmed that all implementations produce biologically consistent results and that UMAP and t-SNE achieve substantially higher embedding quality scores than PCA. These contributions address a clear gap in the existing metabolomics software landscape and significantly extend the analytical capabilities available to users of both specmine and WebSpecmine. The extended versions are publicly available at https://github.com/PedroFontao/Projeto-Bioinf. It should be noted that the new WebSpecmine modules currently operate as a standalone extension and are not yet fully integrated into the upstream specmine R package; full integration is planned as part of future development.

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